Microdialysis Evaluation of Blood Flow, Lactate, and Glucose in Active Trigger Points of Tension-Type Headache Patients and Latent Trigger Points in Healthy Muscle

Arch Phys Med Rehabil Arch Phys Med Rehabil 319 nihpa 2985158R Archives of physical medicine and rehabilitation 0003-9993 1532-821X pmc-is-collection-domain yes pmc-collection-title NIHPA Author Manuscripts PMC13421107 PMC13421107.1 13421107 13421107 NIHMS2195535 41033570 10.1016/j.apmr.2025.09.019 NIHMS2195535 NIHPA2195535 1 Article Microdialysis Evaluation of Blood Flow, Lactate, and Glucose in Active Trigger Points of Tension-Type Headache Patients and Latent Trigger Points in Healthy Muscle Moraska Albert F. PhD 1 * Wang Yue PhD 2 Hickner Robert C. PhD 3 Lynn Rachael Rzasa MD 4 Hebert Jonathan MS 5 Kohrt Wendy M. PhD 6 1 College of Nursing, University of Colorado Anschutz Medical Campus, Aurora, CO, 80045 USA 2 Department of Biostatistics and Informatics, University of Colorado Anschutz Medical Campus, Aurora, CO, 80045 USA 3 Department of Health, Nutrition and Food Sciences and the Institute of Sports Sciences and Medicine, College of Education, Health and Human Sciences, Florida State University, Tallahassee, FL 32306, USA 4 Department of Anesthesiology, School of Medicine, University of Colorado Anschutz Medical Campus, Aurora, CO, 80045 USA 5 Pivot Point Integrative Massage, LLC, Denver, CO, 80209 USA 6 Department of Medicine, University of Colorado Anschutz Medical Campus, Aurora, CO, 80045 USA and VA Eastern Colorado Health Care System, Geriatric Research, Education, and Clinical Center, Aurora, CO 80045 USA Author Contributions. AM. and R.C.H.: contributed to study conceptualization and design, data collection, interpretation of data, and manuscript preparation. Y.W. contributed to the statistical analysis, data interpretation, and manuscript preparation; R.R-L, J.H. and W.M.K. contributed substantially to the study conceptualization, interpretation of data, and manuscript preparation. All authors have read and agreed to the published version of the manuscript. * Corresponding author: Albert.Moraska@CUAnschutz.edu No conflict of interest, financial or otherwise, is declared by the authors. 5 2026 29 9 2025 107 5 518667 914 923 21 07 2026 30 07 2026 31 07 2026 31 07 2026 Objective. To assess nutritive blood flow, lactate, and glucose within active and latent TrPs compared to non-TrP muscle. Design. Case-controlled observational study. Setting. A University-based Clinical and Translational Research Center (CTRC). Participants. Active TrPs were identified through palpation of the upper trapezius muscle of individuals with episodic or chronic tension-type headache (TTH). Latent TrPs and muscle without TrPs were identified in age- and sex-matched subjects without a history of headache. Interventions. Not applicable. Main Outcome Measures. Nutritive blood flow, and interstitial lactate and glucose concentrations were determined at 1220-minute intervals over 160 minutes through microdialysis sampling of interstitial fluid. Results. A total of 64 individuals participated in the study. Subjects were 33.8±10.8y, 64.1% female, and 73.4% white, with no statistical differences between active, latent, and no TrP groups on demographic or baseline variables (p 0.05). Subjects with TTH reported a median of 4 headaches/month and had experienced headaches for a median of 120 months. Active TrPs exhibit higher nutritive blood flow than anatomically matched no TrP muscle (p=0.026) but were not different from latent TrPs (p=0.554). There were no global differences between the three groups for lactate (p=0.329) or glucose (p=0.231). Conclusions. The present study provides in vivo evidence of higher nutritive blood flow in active TrPs from individuals with TTH but no difference in lactate and glucose compared to non-TrP muscle. Our data does not support the TrP as a tightly contracted, ischemic region within skeletal muscle. Interventions designed to relax muscle contraction or increase blood flow may have limited benefits in alleviating active trigger points. Myofascial myofascial pain syndrome skeletal muscle metabolism pain pressure-pain threshold humans casecontrol study pmc-status-qastatus 0 pmc-status-live yes pmc-status-embargo no pmc-status-released yes pmc-prop-open-access no pmc-prop-olf no pmc-prop-manuscript yes pmc-prop-legally-suppressed no pmc-prop-has-pdf yes pmc-prop-has-supplement yes pmc-prop-pdf-only no pmc-prop-suppress-copyright no pmc-prop-is-real-version no pmc-prop-is-scanned-article no pmc-prop-preprint no pmc-prop-in-epmc yes